

V-KEMS

The Virtual Forum for Knowledge Exchange in the Mathematical Sciences

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KE Hub Seminar Series

24 September 2026

School of Computing Science, University of Glasgow



Whitby, 15 March



Cardiff, 18 March



Every UK phone, 24 March

V-KEMS: set up in March 2020

A collaboration between

- Isaac Newton Institute
- Newton Gateway to Mathematics
- International Centre for Mathematical Sciences
- KTN, now Innovate UK Business Connect

First VSG: 20–23 April

2 industrial problems, 30 people

Second VSG: 29–30 April

50 people, and a 50-page report to government two days later

Virtual was both required and beneficial

Our June 2020 operations manual noted that being online meant more people could take part:

- people with caring responsibilities
- people with unavoidable work commitments
- people who could not travel
- domain experts who were only needed for part of the time

Online activities were part of our founding, and remain part of our focus.

Why am I giving this talk?

I want two things from you

If you are an academic

Come and take part.

If you are not

Bring us a problem.

Let me show you why both are worth doing.

How a virtual study group works

- **Day 1:** each problem owner presents their problem (20 minutes, plus questions)
- Participants pick which problem they want to work on
- Teams usually split up into smaller groups of 4-5
- Short progress updates each morning, and a catch-up at the end of the day
- **Last day:** each team presents what they've done, and afterwards writes a report

There's (ideally) funding available to keep working on problems afterwards.

What it costs you

Problem owner

Some time in advance preparing the problem and any data - we can help with this.

Twenty minutes of presentation, be reachable during the three days, and attend final presentations.

Participant

Be comfortable with "mutually assured ignorance", and willing to engage.

You do not have to commit the whole week, but more time is better.

Come for the part where you are useful: this might be most of it! Some time after the event will also be needed.

When a VSG might be the wrong tool

- The problem cannot be discussed openly
- The data cannot be shared in any form, or does not exist yet
- You need a guaranteed answer by a fixed date
- It is not really amenable to mathematical formulation

We can usually quickly help a problem owner figure out what their best use of a VSG is before they commit anything.

I want to tell you about some of our past activities

- Pandemic crisis era: unlocking higher education
- Pandemic-adjacent: cardiovascular waiting lists
- Post-crisis: future of the high street
- More recent: archaeology

Unlocking Higher Education Spaces

June 2020 | Newton Gateway

How do you reopen a campus?

Universities got

a briefing document and a working paper,
while they were deciding how to reopen

The mathematicians got

an interesting new problem - and some of
the discussions fed into later work on
transmission in student populations

...and it led to a paper

Discussions at the study group carried on, and turned into this:

ROYAL SOCIETY
OPEN SCIENCE

royalsocietypublishing.org/journal/rsos

Research



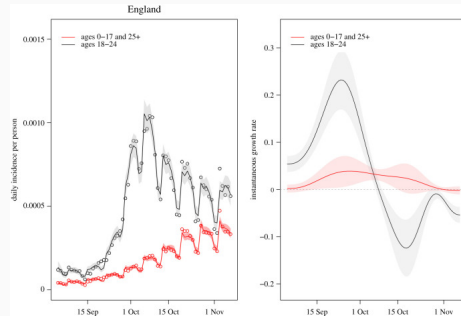
Cite this article: Enright J *et al.* 2021 SARS-CoV-2 infection in UK university students: lessons from September–December 2020 and modelling insights for future student return. *R. Soc. Open Sci.* 8: 210310. <https://doi.org/10.1098/rsos.210310>

Received: 23 February 2021

Accepted: 16 July 2021

SARS-CoV-2 infection in UK university students: lessons from September–December 2020 and modelling insights for future student return

Jessica Enright^{1,†}, Edward M. Hill^{2,3,†}, Helena B. Stage^{3,4}, Kirsty J. Bolton⁵, Emily J. Nixon^{3,6}, Emma L. Fairbanks^{5,7}, Maria L. Tang^{3,8}, Ellen Brooks-Pollock^{3,9}, Louise Dyson^{2,3}, Chris J. Budd¹⁰, Rebecca B. Hoyle¹¹, Lars Schewe¹², Julia R. Gog^{3,8} and Michael J. Tildesley^{2,3}



Cases in 18–24 year olds (black) and everyone else (red), England, autumn 2020

Enright, Hill *et al.* (2021) *R. Soc. Open Sci.* 8:210310, CC BY 4.0

Cardiovascular waiting lists

February 2021 | NHS, Newton Gateway

How has COVID affected cardiovascular waiting lists, and what can be done about it?

The NHS got

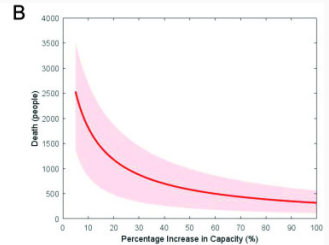
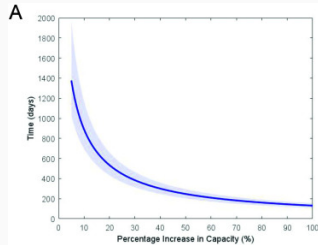
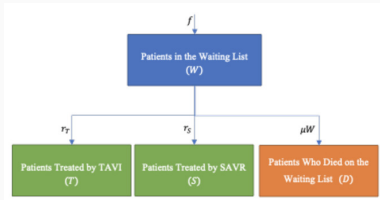
models of the referral, diagnosis and treatment pathway, to help plan how to bring waiting lists down

The mathematicians got

two papers in *BMJ Open* (and I got to be on a podcast!)

What did the models say?

One example: waiting lists for aortic stenosis treatment



Time to clear the backlog (left) and deaths on the waiting list (right), against the increase in treatment capacity

Even a modest increase in capacity makes a big difference, both to how long the backlog takes to clear and to how many people die waiting.

The Future of the High Street

May 2024 | Innovate UK Business Connect

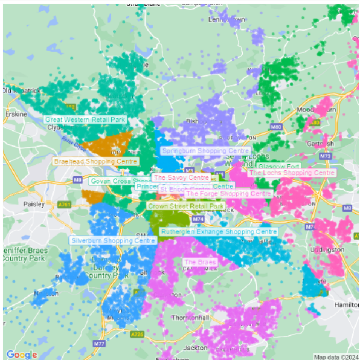


High Street, Selkirk, 2022. Richard Webb, CC BY-SA 2.0, via Geograph

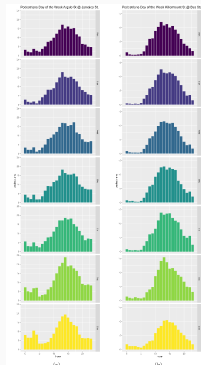
- What does out-of-town shopping do to city centres?
- What are the benefits of more people living in Scottish city centres? (from the Scottish Cities Alliance)

Two reports came out of this one.

Where do people in Glasgow shop?



Glasgow, split up by nearest shopping area



Pedestrians by hour, for each day of the week: Argyle St (left) and Killermont St (right), from Glasgow's open CCTV counts

From the V-KEMS report on this study group. Map data © 2024 Google.

Mathematics in Archaeology and Museum Collections

September 2025 | Institute for Mathematical Innovation, Fitzwilliam Museum

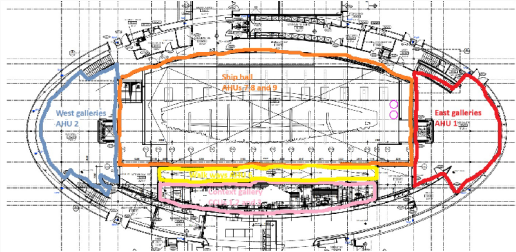


The hull of the Mary Rose. Bob Harvey, CC BY-SA 2.0, via Geograph

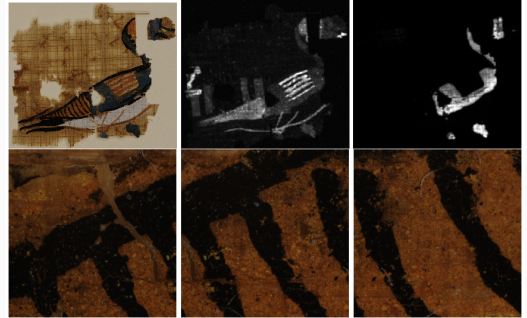
- Energy use at the Mary Rose Museum
- Inheritance patterns in Roman Britain, from ancient DNA
- Image registration and stitching for cultural heritage

Would you have guessed that
the Mary Rose Museum
had a maths problem?

A Tudor warship and an Egyptian papyrus



The Mary Rose has to be kept at 18–20 °C and 50–58% humidity.
What drives the museum's energy use?



Fragments of the Book of the Dead of Ramose (c. 1300 BC)
under different imaging methods - how do you line the images up?

Figures from the study group reports. Right: © Fitzwilliam Museum, University of Cambridge.

Where are we now?

- More than 20 virtual study groups since 2020
- On water, food security, justice, higher education funding, museums, and lots more
- KE Team of the Year at the PraxisAuril awards in 2021
- Next up: policing problems, with the NPCC, in December

We started as a COVID response, but we've kept going!

Coming up: policing problems

1–3 December 2026 | online, three days

With the National Police Chiefs' Council and the Policing Academic Centres of Excellence, based on the NPCC's *Policing Problem Book*

Three challenges (details still provisional):

- How does the risk of crime escalate with age, after petty crime in youth?
- Measuring and reducing systemic bias - e.g. in stop and search
- Logistics: drone routing, airspace coordination, where to put assets

If you'd like to take part, keep an eye on vkemsuk.org!

What might we do in the future?

- **Virtual study groups** - still our main activity
 - trialling: coupling these with regular online working groups
- **Hybrid study groups** - some people in the room, others online
- **Webinars and scoping meetings** - shorter, to work out whether a problem is a good fit
- **Ongoing collaboration with the KE ecosystem** - e.g. KE Hub, Academy

Please get involved!

Academics

Sign up for a study group - and come for as much of it as you can. Early career colleagues especially welcome!

Everyone else

Get in touch with your problems, even if they're not fully formed yet.

vkemsuk.org

Want to join us? We're recruiting!

We're looking for an **early career researcher** to join the V-KEMS organising committee, leading on **communications and marketing**.

- Build your network and get committee leadership experience
- Help connect mathematics with real-world problems
- Voluntary and unpaid, like all our committee roles

Deadline: 5pm, Wednesday 14 October 2026

Details on the Events & Opportunities page at vkemsuk.org

Questions